# Class-Imbalanced Learning via Multitask Multiobjective Feature Selection

Ruwang Jiao, *Senior Member, IEEE*, Naoki Masuyama, *Member, IEEE*, and Yusuke Nojima, *Member, IEEE*

***Abstract*—Class-imbalanced and high-dimensional data pose significant challenges in machine learning, often leading to learning models that underperform on minority classes and struggle with the curse of dimensionality, respectively, and these two issues are inherently interconnected. This article simultaneously addresses both class imbalance and high dimensionality through a multitask multiobjective feature selection method. To be specific, the proposed method establishes two related yet distinct feature selection problem formulations for a given task, each focusing on different aspects of overall and minority classification performance. By selecting important features from the original imbalanced dataset, the proposed method avoids the need to artificially equalize class distribution during training. In addition, the cross-task genetic transfer operator enables the reuse of valuable feature subsets evolved from separate searches, thereby improving both minority class accuracy and overall model performance. Experimental results on 17 class-imbalanced datasets show that the proposed method achieves better results in terms of overall classification performance and minority class accuracy than the compared state-of-the-art methods, particularly on the high-dimensional datasets.**

***Impact Statement*—The high-dimensional class-imbalanced data present significant challenges in machine learning, especially in critical fields such as healthcare, finance, and bioinformatics, where addressing the curse of dimensionality and ensuring accurate predictions for minority classes are essential. This article introduces a novel multitask multiobjective feature selection method that effectively addresses both issues, enhancing classification performance without requiring artificial class balancing. The proposed method improves minority class accuracy while maintaining overall model effectiveness. Moreover, the feature subsets obtained by the proposed method can all achieve similar or even lower classification error rates than using all features, while significantly reducing data dimensionality. This research has the potential to drive more accurate, efficient models, thereby reducing costs, increasing reliability, and supporting better decision-making in vital industries.**

***Index Terms*—Class-imbalanced learning, evolutionary feature selection, multiobjective learning, multitask learning.**

Received 27 May 2025; revised 3 November 2025 and 25 January 2026; accepted 9 March 2026. Date of publication 13 March 2026; date of current version 31 August 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62406212, in part by the Science and Technology Program of Jiangsu Province under Grant BK20251895, in part by Suzhou Planning Project of Science and Technology under Grant ZXP2025060, in part by China Postdoctoral Science Foundation under Grant 2025M781499, in part by the Fundamental Research Funds for the Central Universities, JLU, and in part by JSPS KAK-ENHI under Grant 23K24920 and Grant 24KF0254. This article was recommended for publication by Associate Editor Bing Xue upon evaluation of the reviewers' comments. *(Corresponding author: Ruwang Jiao.)*

Ruwang Jiao is with the School of Future Science and Engineering, and the Key Laboratory of General Artificial Intelligence and Large Models in Provincial Universities, Soochow University, Suzhou 215222, China, and also with the Key Laboratory of Symbolic Computation and Knowledge Engineering of Ministry of Education, Jilin University, Changchun 130012, China (e-mail: ruwangjiao@gmail.com).

Naoki Masuyama and Yusuke Nojima are with the Department of Core Informatics, Graduate School of Informatics, Osaka Metropolitan University, Osaka 599-8531, Japan (e-mail: masuyama@omu.ac.jp; nojima@omu.ac.jp).

This article has supplementary downloadable material available at https://doi.org/10.1109/TAI.2026.3673680, provided by the authors.

Digital Object Identifier 10.1109/TAI.2026.3673680

## I. Introduction

LEARNING from imbalanced data is a prevalent yet challenging task in machine learning and data mining, where the number of samples in the majority class far exceeds the number of samples in the minority class [1], [2], [3]. This imbalance leads to learning models that typically exhibit poor performance on the minority class due to the accuracy-oriented design of standard classification methods and data difficulty factors such as within-class imbalance, class overlapping, and scarce representative samples. Thus, accurate predictions for the minority class, often the class of primary interest, are greatly compromised.

In the era of big data, the challenge of class imbalance is frequently intensified by the high dimensionality of datasets. These two issues are often interrelated [4]. To be specific, high dimensionality can easily lead to overfitting, particularly for minority-class samples that are underrepresented. Conversely, the class imbalance can exacerbate the sparsity of data in high-dimensional spaces, making it difficult for classifiers to identify enough minority-class samples to learn from, ultimately hindering generalization and model performance.

Class-imbalanced learning aims to address the challenges posed by datasets with highly skewed class distributions. Class-imbalanced learning methods can be broadly categorized into two approaches: data-level and algorithm-level [4]. Data-level methods primarily focus on preprocessing to rebalance the data distribution. This often involves resampling techniques that adjust the number of samples for each class, aiming to create a more equitable representation. By contrast, algorithm-level approaches modify the learning algorithms themselves to reduce bias, ensuring that the learning model is more likely to favor minority-class samples during training.

To address high-dimensional data, feature selection is a primary technique for dimensionality reduction [5]. Its goal is to eliminate as many irrelevant, noisy, and redundant features as possible without compromising and potentially even enhancing classification performance. Irrelevant, noisy, and redundant features can severely impact classification performance for several reasons: 1) irrelevant features provide useless information for predicting class labels, and their presence can greatly diminish model accuracy; 2) noisy features introduce misleading data, which negatively affects classification results; and 3) redundant features duplicate information already present in other features, increasing computation time without contributing to model improvement.

Note that although deep learning has achieved remarkable success in many domains, feature selection still has some distinct advantages [6]. Deep learning models are often regarded as black boxes, whereas feature selection produces compact and interpretable feature subsets, facilitating a better understanding of feature importance and interactions. Moreover, feature selection generally requires less training data and fewer computational resources, making it more accessible than deep learning approaches. While deep learning excels in areas such as computer vision and natural language processing, it often performs poorly on heterogeneous tabular data with small sample sizes and extremely high dimensionality [7], which is the focus of this work.

While resampling techniques are commonly used to balance class distributions by adjusting the number of samples from different classes, feature selection provides a powerful alternative for addressing class imbalance [3]. By identifying the most discriminative features for each class, feature selection accounts for the fact that different features contribute unequally to the classification of various classes [8]. In addition, one key advantage of feature selection is that it preserves the original data values and the number of training samples, maintaining the data's inherent semantics while reducing its dimensionality. By focusing the model on the most informative features, feature selection enhances its ability to capture underlying patterns, even in the presence of significant class imbalance.

However, feature selection is a challenging combinatorial optimization task for several reasons [6], [9]. First, the search space increases exponentially with the number of features, making it an nondeterministic polynomial time (NP)-hard problem. Second, feature selection is influenced by complex feature relationships. For instance, redundant features convey similar information and do not contribute to improved classification performance, while complementary features, when combined, can enhance class label discrimination. Third, the optimal number of features to be selected is typically unknown and problem-dependent, making it difficult for feature selection methods to predefine the appropriate number of features to select. Fourth, feature selection is inherently a multiobjective task, where the two main objectives often conflict with one another, e.g., removing relevant features can reduce classification accuracy. As a result, it is difficult to find an optimal subset of features that strikes a balance between classification performance and the number of features selected.

Evolutionary computation (EC) methods [10], [11] have gained popularity for both feature selection [6], [12] and class-imbalanced learning [13] tasks due to their simplicity and versatility. EC methods have global search ability, which effectively explores the solution space and reduces the likelihood of getting trapped in local optima. In addition, they do not rely on gradient information from the optimization functions, which makes them particularly well-suited for problems involving discrete encodings and black-box objective functions. More importantly, their population-based characteristic facilitates the generation of a diverse set of tradeoff solutions, which is especially beneficial for optimizing multiple conflicting objectives. Unlike traditional feature selection methods that require a predefined number of features to be selected, multiobjective EC methods can automatically generate a set of tradeoff feature subsets, balancing the number of selected features and classification accuracy.

So far, limited attention has been given to addressing both class imbalance and high-dimensional data issues simultaneously. Furthermore, most approaches for class-imbalanced learning prioritize improving the performance of the minority class, often at the expense of degrading the performance of other classes. To address this, in this article, we propose CIL-MTFS, a novel multitask multiobjective feature selection method designed for class-imbalanced learning task. By leveraging the strengths of EC in global feature subset search and multiobjective optimization, CIL-MTFS addresses the challenge of selecting informative features while considering both overall and minority class classification performance. Specifically, CIL-MTFS formulates two interrelated feature selection problems for a given task, each targeting different aspects of classification performance. By selecting key features directly from the original imbalanced dataset, CIL-MTFS eliminates the need for artificial class balancing during training, preserving the data's intrinsic characteristics while reducing dimensionality. Furthermore, the cross-task genetic transfer operator in CIL-MTFS allows for the reuse of valuable feature subsets evolved across different searches, enhancing accuracy for minority classes and improving overall model performance.

The main contributions of this article can be summarized as follows.

1) Propose to formulate the feature selection task in class-imbalanced learning as a multitask problem, where each task has a distinct focus. The first task is a constrained multiobjective optimization problem that aims to optimize both overall classification performance and the number of selected features. Meanwhile, the constraint is applied to eliminate feature subsets that select very few features but demonstrate poor accuracy. While these subsets could be nondominated, they are practically undesirable and thus excluded. The second task specifically targets improving the classification performance of the minority class. The proposed multitask method can simultaneously reduce data dimensionality and enhance classification performance for high-dimensional and class-imbalanced data.

2) Design a knowledge-sharing mechanism to facilitate the transfer of informative and generalizable features between tasks. This mechanism enables the effective reuse of feature subsets evolved through cross-task genetic transfer, which is helpful to enhance both overall classification performance and the performance of the minority class.

Note that the main differences between our proposed method and existing multitask EAs lie in the goal of the multitask framework and the way multiple tasks are defined and formulated. Existing methods, such as Bi-Search [14] and multitask PSO/CSO [15], [16], typically define multiple tasks *based on the number of selected features*, decomposing a high-dimensional feature selection problem into several lower dimensional ones and transferring knowledge between them to accelerate convergence. By contrast, our method defines multiple tasks *at the problem formulation level* rather than by feature dimensionality: the first task is a constrained multiobjective optimization problem that jointly optimizes classification accuracy and feature sparsity, while the second task focuses on improving minority-class performance.

The remainder of this article is organized as follows. Section II reviews the related work about class-imbalanced learning, multiobjective, and multitask feature selection. Section III presents the details of the proposed CIL-MTFS method. Section IV outlines the experimental design. Section V provides discussions and analysis. Finally, the article concludes with a summary in Section VI.

## II. Related Work

This section provides a brief review of related work pertinent to this article, covering data-level and algorithmic approaches for class-imbalanced learning, as well as methods for multiobjective and multitask feature selection.

### *A. Data-Level Methods for Class-Imbalanced Learning*

Data-level class-imbalanced learning methods focus on manipulating the data distribution to balance the class distribution, thereby reducing bias in the learning process. Techniques in this category include undersampling, oversampling, and hybrid sampling.

Undersampling methods [17], [18] involve removing some samples from the majority class based on specific criteria, aiming to make the number of samples in the majority class roughly equal to that in the minority class. However, undersampling can lead to the loss of valuable majority-class data, which might degrade the learning performance on the reduced dataset [19].

In contrast to undersampling, the primary goal of oversampling is to increase the number of samples in the minority class to match or approximate the number of samples in the majority class. This can be achieved by either replicating existing samples from the minority class for multiple uses or by generating synthetic samples [20], [21], [22] to augment the minority class. While this approach helps address class imbalance, simply replicating minority class instances can lead to overfitting, as it might reinforce the noise or redundancies in the data.

Hybrid sampling combines the strengths of both undersampling and oversampling methods to address the class imbalance issue [23]. It aims to balance the dataset by strategically reducing the number of majority-class samples and simultaneously increasing the number of minority-class samples.

### *B. Algorithm-Level Methods for Class-Imbalanced Learning*

Algorithm-level class-imbalanced learning methods involve modifying the learning algorithms themselves to handle imbalanced data more effectively [24]. Techniques in this category include cost-sensitive learning, ensemble learning, one-class classification, and kernel modification methods.

In cost-sensitive learning [25], misclassification costs are adjusted to penalize errors in the minority class more heavily, thus improving the classification model's sensitivity to minority-class samples. Ensemble learning techniques [26], [27] such as bagging and boosting combine multiple weak learners to create a stronger model, improving performance on imbalanced datasets. Boosting focuses on misclassified samples, prioritizing the minority class, while bagging reduces overfitting by averaging predictions from models trained on different data subsets.

One-class classification [28] focuses on modeling only the majority class. The model learns the characteristics of the normal class and identifies samples that deviate from it, treating anomalies as potential minority class examples. This method is particularly useful when data for the minority class is scarce, such as in fraud detection or rare disease diagnosis. Kernel modification methods [29] alter the internal representations used by learning algorithms, to better capture the characteristics and complexities of the minority class.

### *C. Multiobjective Feature Selection for Class-Imbalanced Learning*

The data-level resampling techniques and algorithm-level methods discussed above might not suffice to tackle *high-dimensional* class imbalance problems. Feature selection offers an alternative solution, addressing both class imbalance and high dimensionality simultaneously [30].

Multiobjective feature selection [31], [32], [33], [34] is an approach that simultaneously optimizes multiple objectives in the process of selecting useful features from a dataset, such as maximizing accuracy and minimizing the number of selected features. This process generates a set of optimal solutions (e.g., feature subsets), known as the Pareto front, where each solution represents a different tradeoff among the objectives. By exploring these tradeoffs, it provides decision makers with a set of feature subsets to choose from, allowing them to select the one that best aligns with their specific interests.

When applying multiobjective feature selection to class-imbalanced learning tasks, a common strategy is to carefully define the objective functions to ensure that the selected features enhance learning accuracy for the minority class. An improved multiobjective fireworks algorithm [35] is proposed to address imbalanced classification problems. This approach treats feature selection as a multimodal multiobjective optimization problem, aiming to identify multiple equivalent feature subsets within the search space. A Jaccard similarity-based many-objective feature selection method [36] is employed to address the challenges of high dimensionality and imbalanced data simultaneously. This method treats feature selection as a many-objective task, optimizing precision, recall, specificity, and feature subset size. By integrating Jaccard similarity into the population initialization, offspring generation, and environmental selection, it enhances population diversity and prevents duplicate feature subsets.

### D. Evolutionary Multitask Feature Selection

Multitask learning [37] is a form of transfer learning where multiple related tasks are learned simultaneously, sharing information and representations across tasks to enhance overall performance. By leveraging the commonalities and differences among these related but distinct tasks, multitask learning can achieve better generalization and efficiency compared to learning each task independently. This approach is especially advantageous when tasks have limited data, as the shared knowledge among tasks can significantly improve learning performance for individual tasks.

The combination of multitask learning with EC to boost the performance of multiple optimization or learning tasks simultaneously has garnered widespread attention [38], [39]. A number of evolutionary multitask feature selection methods have been developed to leverage the benefits of multitask learning, enabling the simultaneous resolution of multiple feature selection tasks [15], [16], [40], [41]. For example, a PSO-based multitask learning method [42] is used to address the high-dimensional feature selection task by creating two subpopulations. The first enhances knowledge transfer by integrating the other's optimal solutions. The second uses this shared knowledge to refine its search for valuable features, improving the algorithm's efficiency and effectiveness. The multiform feature selection method [43] constructs auxiliary tasks based on the feature subset distribution of the source task within the objective space, enabling the search for feature subsets that exhibit improved convergence and diversity in a multitask environment. The bi-search evolutionary feature selection method [14] integrates two distinct tasks, i.e., forward and backward search, to enhance the search capability within large-scale search spaces, particularly for high-dimensional data.

### E. Rationale and Discussions

Although several studies on multitask feature selection exist, they primarily aim to enhance diversity [43] or accelerate convergence [44] in the target multiobjective task by leveraging auxiliary single-objective tasks, rather than focusing on class-imbalanced data. As discussed above, data-level and algorithm-level methods can alleviate the class imbalance issue to some extent, but they often fail to tackle the high-dimensionality challenge, particularly when the number of features greatly exceeds the number of training samples. This disparity can lead to a sparse feature space, increasing the risk of overfitting. In classification tasks, certain features are strongly correlated with specific class labels, making them critical for distinguishing between classes. Feature selection can identify these important features, particularly those relevant to the minority class, thereby improving its classification accuracy [45]. However, existing feature selection methods primarily focus on enhancing minority class performance, which might inadvertently degrade the performance of the majority class.

To address this, the next section will introduce the proposed CIL-MTFS method, which constructs two related tasks, each with its own emphasis. For example, the first task focuses on maximizing overall classification accuracy, while the second task prioritizes the accuracy of the minority class. This dual-task approach facilitates the effective reuse of feature subsets through cross-task genetic transfer, thereby improving both minority class accuracy and overall classification performance.

## III. Proposed CIL-MTFS Method

### A. Task Generation

CIL-MTFS defines two tasks. The first task is formulated as a constrained multiobjective optimization problem that jointly optimizes overall accuracy and feature sparsity, while applying a constraint to exclude feature subsets whose training accuracy is worse than that achieved using all features. The second task focuses on enhancing the classification performance of minority classes.

*1) Task $\mathcal{T}_1$:* It focuses on the overall classification performance. Minimizing the classification error rate $f_{\text{err}}(\boldsymbol{x})$ and the selected feature ratio $f_{\text{ratio}}(\boldsymbol{x})$ can be considered as a constrained multiobjective optimization problem

$$\begin{aligned} \min \quad & \boldsymbol{F}^{(1)}(\boldsymbol{x}) = (f_{\text{err}}(\boldsymbol{x}), f_{\text{ratio}}(\boldsymbol{x})) \\ \text{s.t.} \quad & g(\boldsymbol{x}) = f_{\text{err}}(\boldsymbol{x}) - f_{\text{err}}(\boldsymbol{x}_{\text{full}}) \leq 0 \\ \text{where} \quad & \boldsymbol{x} = (x_1, \ldots, x_D) \in \Omega = \{0,1\}^D \end{aligned} \tag{1}$$

where $D$ is the total number of features in a dataset. $\boldsymbol{x}$ is a binary vector that represents a solution, i.e., a feature subset, where $x_i = 1$ and $x_i = 0$ mean that the $i$th feature is selected and discarded, respectively.

The first objective $f_{\text{err}}(\boldsymbol{x})$ is the balanced classification error rate

$$f_{\text{err}}(\boldsymbol{x}) = 1 - \frac{1}{c}\sum_{i=1}^{c} \frac{\text{TP}_i}{|S_i|} \tag{2}$$

where $c$ is the number of classes of a dataset. $\text{TP}_i$ represents the number of correctly identified samples in class $i$, and $|S_i|$ denotes the sample size of class $i$.

The second objective $f_{\text{ratio}}(\boldsymbol{x})$ is the ratio of the number of selected features to the total number of features

$$f_{\text{ratio}}(\boldsymbol{x}) = \frac{1}{D}\sum_{i=1}^{D} x_i. \tag{3}$$

The constraint $g(\boldsymbol{x})$ is used to ensure that the training error rate after feature selection is less than or equal to that using all features (i.e., the full set $\boldsymbol{x}_{\text{full}}$). It is also employed to prevent

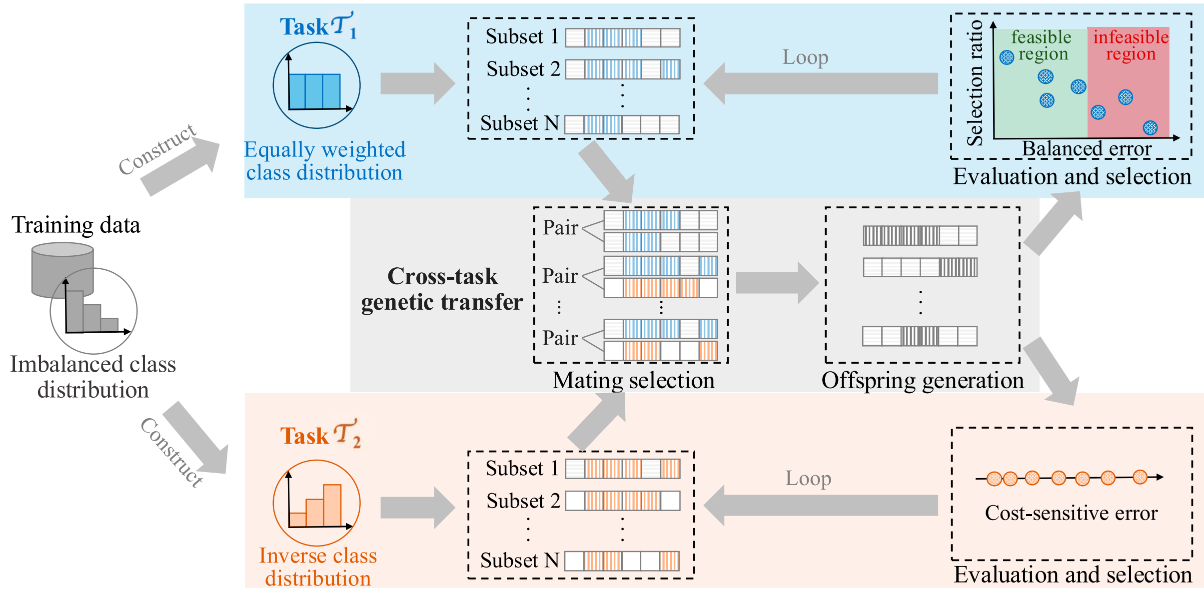


Fig. 1. Diagram of CIL-MTFS.

the feature subsets that select a very small number of features but exhibit a large classification error rate, even if such solutions might be nondominated ones.

*2) Task $\mathcal{T}_2$:* It aims to prioritize the performance of the minority class, which uses a cost-sensitive objective function

$$\begin{aligned} \min \quad & \boldsymbol{F}^{(2)}(\boldsymbol{x}) = 1 - \sum_{i=1}^{c} w_i \frac{\mathrm{TP}_i}{|S_i|} \\ \text{where} \quad & \boldsymbol{x} = (x_1, \ldots, x_D) \in \Omega = \{0,1\}^D \\ & \sum_{j=1}^{c} w_j = 1 \end{aligned} \tag{4}$$

where $w_i$ is the weight for the $i$th class, which is determined by two key factors.

a) *Data perspective*: The smaller the number of samples in a class, the larger the weight would be assigned to it.

b) *Algorithm perspective*: The lower the accuracy for the minority class in an algorithm, the higher the weight that should be assigned to it.

To achieve the above goals, (5) is used to define the weight of the $i$th class

$$\boldsymbol{w} = \begin{cases} \boldsymbol{w}^{\mathrm{dat}}, & \text{if } \max\limits_{i\in[1:c]} w_i^{\mathrm{dat}} \geq \max\limits_{i\in[1:c]} w_i^{\mathrm{alg}} \\ \boldsymbol{w}^{\mathrm{alg}}, & \text{else} \end{cases}$$

where

$$w_i^{\mathrm{dat}} = \frac{|S_i|^{-1}}{\sum_{k=1}^{c} |S_k|^{-1}},$$

$$w_i^{\mathrm{alg}} = \frac{\frac{1}{N}\sum_{j=1}^{N}\left(1 - \frac{\mathrm{TP}_i(\boldsymbol{x}_j)}{|S_i|}\right)}{\sum_{k=1}^{c}\left(\frac{1}{N}\sum_{j=1}^{N}\left(1 - \frac{\mathrm{TP}_k(\boldsymbol{x}_j)}{|S_k|}\right)\right)} \tag{5}$$

where $N$ is the population size. Equation (5) assigns class weights based on either sample size or average classification error in the current population. This ensures that even classes with many samples but high error rates receive greater attention. Specifically, the weight is the maximum of two ratios: one reflecting class imbalance and the other reflecting relative error, allowing the model to dynamically prioritize underrepresented or poorly classified classes.

## B. Overall CIL-MTFS Framework

The overall framework of the proposed CIL-MTFS method is presented in Fig. 1 and its pseudocode is shown in Algorithm 1, with the key focus on how to leverage the two tasks defined in (1) and (4) for class-imbalanced learning. CIL-MTFS begins by evaluating the classification performance using all features (i.e., $\boldsymbol{x}_{\mathrm{full}}$), which is then used as the constraint boundary for (1). Subsequently, a population is randomly generated and evaluated according to the objective functions defined in (1) and (4), designated as $\mathcal{P}_1$ and $\mathcal{P}_2$, respectively. For each initial solution $\boldsymbol{x}$, the number of selected features is randomly determined, with an upper bound of three times the population size, i.e., $|\boldsymbol{x}| = \mathrm{rand}(1, \min(D, 3N))$ [46].

In each iteration, two mating populations, $MP_1$ and $MP_2$, are generated by selecting solutions from $\mathcal{P}_1$ and $\mathcal{P}_2$, respectively. To be specific, binary tournament selection is used for this purpose: for $MP_1$, the selection is based on constrained nondominated ranking, where feasible solutions are always preferred over infeasible ones, and among feasible solutions, the one with the lower nondominated rank is favored. For $MP_2$, the selection is guided by the cost-sensitive error rate defined in (4). An offspring population $\mathcal{O}$ is then generated through the offspring generation operator, facilitating knowledge transfer and sharing between tasks $\mathcal{T}_1$ and $\mathcal{T}_2$, as detailed in Algorithm 2. The offspring population $\mathcal{O}$ is evaluated for $\mathcal{T}_1$ based on (1) and for $\mathcal{T}_2$ based on (4). Environmental selection is subsequently applied separately to update $\mathcal{P}_1$ and $\mathcal{P}_2$ by selecting from their respective unions with $\mathcal{O}$. This process continues until a stopping criterion is met, i.e., a predefined maximum number of function evaluations, after which the algorithm returns the feasible nondominated solutions in $\mathcal{P}_1$.

**Algorithm 1:** CIL-MTFS.

**Input:** The population size $N$.
**Output:** The feasible non-dominated solutions in $\mathcal{P}_1$.
1 Evaluate the classification error rate using all features, i.e., $f_{err}(\boldsymbol{x}_{full})$;
2 Initialize a population and evaluate it according to Eq. (1) and Eq. (4), and regard it as $\mathcal{P}_1$ and $\mathcal{P}_2$, respectively;
3 **while** the stopping criterion is not met **do**
4 $MP_1 \leftarrow$ Binary tournament selection is performed on $\mathcal{P}_1$ based on constrained non-dominated ranking; /* Mating selection of $\mathcal{P}_1$ */
5 $MP_2 \leftarrow$ Binary tournament selection is performed on $\mathcal{P}_2$ based on cost-sensitive error rate; /* Mating selection of $\mathcal{P}_2$ */
6 $\mathcal{O} \leftarrow$ Cross-task genetic transfer ($MP_1$, $MP_2$); /* Algorithm 2 */
7 Evaluate $\mathcal{O}$ for $\mathcal{T}_1$ and $\mathcal{T}_2$ according to Eq. (1) and Eq. (4), respectively;
8 $\mathcal{P}_1 \leftarrow$ Perform environmental selection for $\mathcal{P}_1 \bigcup \mathcal{O}$ based on the constrained dominance criterion;
9 $\mathcal{P}_2 \leftarrow$Perform environmental selection for $\mathcal{P}_2 \bigcup \mathcal{O}$ based on the cost-sensitive classification error rate;
10 **end**
11 Return the feasible non-dominated solutions in $\mathcal{P}_1$.

**Algorithm 2:** Cross-Task Genetic Transfer.

**Input:** Mating populations $MP_1$ and $MP_2$ for tasks $\mathcal{T}_1$ and $\mathcal{T}_2$, respectively. The population size $N$. Number of training samples $|S_i|$ for class $i$, and $i = 1, \cdots, c$.
**Output:** Offspring population $\mathcal{O}$.

$k \leftarrow \arg\min_{i\in\{1,\ldots,c\}} (|S_i|)$; /* Identify the minority class */
$\overline{err}_k \leftarrow \frac{1}{N} \sum_{\boldsymbol{x}\in\mathcal{P}_1} (1 - \frac{TP_k(\boldsymbol{x})}{|S_k|})$; /* Average population error rate of the minority class */
$\mathcal{O} = \emptyset$;
**for** $i$=1 to $\frac{N}{2}$ **do**
  **if** $1 - \frac{TP_k(MP_1^i)}{|S_k|} > \overline{err}_k$ **then**
    $MP_1^{i+\frac{N}{2}} \leftarrow MP_2^{i+\frac{N}{2}}$;
  **if** $1 - \frac{TP_k(MP_1^{i+\frac{N}{2}})}{|S_k|} > \overline{err}_k$ **then**
    $MP_1^i \leftarrow MP_2^i$;
**end**
Apply the single-point crossover and bit-flip mutation to generate the offspring population $\mathcal{O}$;
Remove duplicated feature subsets (in the search space) in $\mathcal{O}$;
Return the offspring population $\mathcal{O}$.

Since $\mathcal{T}_1$ and $\mathcal{T}_2$ have different task formulations: $\mathcal{T}_1$ being a constrained multiobjective optimization problem and $\mathcal{T}_2$ a single-objective optimization problem, we employ different environmental selection strategies for each based on their task characteristics. In line 8 of Algorithm 1, for $\mathcal{T}_1$, solutions are first compared based on their degree of constraint violation. If both solutions are feasible, the commonly used nondominated sorting and crowding distance from NSGA-II [47] are applied to select the best $N$ solutions from the union of the parent population $\mathcal{P}_1$ and the offspring population $\mathcal{O}$. Otherwise, the comparison is based on the degree of constraint violation of each solution. Specifically, for any two feature subsets, a feasible feature subset is always preferred over an infeasible one. If both subsets are infeasible, the one with the smaller constraint violation value is considered better. When both feature subsets are feasible, their objective values are compared to determine the better one. For $\mathcal{T}_2$, environmental selection is based solely on the cost-sensitive classification error rate, as defined in (4).

The major novelty of CIL-MTFS lies in its dual-task formulation design and the introduction of cross-task genetic transfer. The core of this transfer is the effective utilization of the two tasks defined in (1) and (4) to address class-imbalanced learning within a multitask framework. In the following subsections, we delve into the cross-task genetic transfer operator, focusing on three critical aspects: when to transfer, what to transfer, and how to transfer.

### C. Evolutionary Multitask Learning

*1) When to Transfer:* Deciding when to transfer is the first question that the proposed CIL-MTFS method needs to address. Transfer should occur when one task's feature subsets can provide valuable insights to improve the other task. In evolutionary computation, mating selection is used to identify and pair solutions based on their fitness. These selected mating solutions are then used to generate offspring through genetic operations. The aim is to combine beneficial traits from both mating parents to produce improved offspring solutions, driving the evolution toward better performance over generations.

In the proposed method, the population $\mathcal{P}_1$ in task $\mathcal{T}_1$ retains feature subsets that consider both the balanced classification error rate and the number of selected features. In contrast, the population $\mathcal{P}_2$ in task $\mathcal{T}_2$ stores feature subsets that achieve smaller classification error rates, with a bias toward the minority class. The transfer in CIL-MTFS is triggered under the following condition: When a mating solution selected from $\mathcal{P}_1$ exhibits a minority class classification error rate that is worse than the average classification error rate of the minority class $k$ in $\mathcal{P}_1$, the second mating solution should be selected from $\mathcal{P}_2$ that demonstrates a good classification error rate for the minority class, indicating that it has identified useful features or patterns that could benefit the minority class performance. Specifically, a solution from task $\mathcal{T}_1$ triggers knowledge transfer from $\mathcal{T}_2$ with a probability proportional to its excess minority-class error, defined as $(1 - (\mathrm{TP}_k/|S_k|) - \overline{\mathrm{err}}_k)$.

It is worth noting that, unlike conventional multitask feature selection methods that typically perform random or uniform knowledge transfer between tasks using a fixed random mating probability (rmp) [15], [40] our approach can *selectively* and *adaptively* transfer knowledge based on class-specific performance discrepancies.

*2) What to Transfer:* Task $\mathcal{T}_1$ emphasizes overall classification performance, while task $\mathcal{T}_2$ focuses on minority class performance. If mating solutions are selected exclusively from the task $\mathcal{T}_1$, the resulting offspring are likely to prioritize overall classification performance. Conversely, if mating solutions are selected only from the task $\mathcal{T}_2$, the offspring are likely to perform well on the minority class but might neglect the performance of other classes. This occurs because solutions in population $\mathcal{P}_2$ tend to select features specifically related to the minority class.

When the feature subsets in task $\mathcal{T}_1$ demonstrate strong overall classification performance but underperform for the minority class, features that might enhance minority class performance should be transferred from task $\mathcal{T}_2$ to the feature subsets in task $\mathcal{T}_1$. Conversely, if the feature subsets in task $\mathcal{T}_2$ excel at classifying the minority class but lag in overall classification performance, features that might improve overall performance should be transferred from task $\mathcal{T}_1$ to task $\mathcal{T}_2$. These conditions ensure that the knowledge exchange is beneficial and targets the specific weaknesses of each task, i.e., boost the performance of both overall classification performance and the minority class.

*3) How to Transfer:* To enhance both overall classification performance and minority class performance, mating solutions for offspring generation must be selected carefully. Since the solutions in $\mathcal{P}_1$ consider the overall classification performance across all classes, we prioritize selecting mating solutions from $\mathcal{P}_1$. However, if a solution from $\mathcal{P}_1$ has a minority class error rate that exceeds the average minority class error rate of the population $\mathcal{P}_1$, its paired mating solution will be replaced by one from $\mathcal{P}_2$ that has demonstrated a better classification error rate for the minority class.

Equation (4) in $\mathcal{T}_2$ does not directly account for the number of selected features. During the mating selection process, if two solutions from $\mathcal{P}_2$ exhibit the same classification error, a secondary comparison based on feature subset size is employed. In this scenario, the solution with fewer selected features is preferred as the mating solution. This ensures that the task $\mathcal{T}_2$ prioritizes the classification performance of the minority class while also implicitly considering feature reduction.

Algorithm 2 presents the pseudocode for the cross-task genetic transfer operator, which is designed to select high-quality mating parents from two tasks and generate promising offspring feature subsets through knowledge transfer. The process begins by identifying the minority class $k$, followed by calculating the average minority class error rate of $\mathcal{P}_1$. As outlined, lines 5 to 8 of Algorithm 2 determine the feature subsets selected as mating parents for the generation of new offspring feature subsets. Once the mating parents are determined, the ordinary offspring generation operators, e.g., single-point crossover and bit-flip mutation, can be applied to produce the offspring population $\mathcal{O}$. Afterward, a duplication-checking operator is applied to $\mathcal{O}$ to remove any duplicated feature subsets that select the same features, retaining only one instance of each to keep the feature subset diversity. The remaining feature subsets are then returned for subsequent objective evaluation.

### D. Analysis of Cross-Task Genetic Transfer

In imbalanced learning, the optimization landscape of the overall error objective $f_{\mathrm{err}}(\boldsymbol{x})$ in Task $\mathcal{T}_1$ is often dominated by majority-class performance, leading the search toward solutions that achieve low total error but poor minority-class accuracy. By contrast, the cost-sensitive objective in Task $\mathcal{T}_2$ introduces a complementary bias toward minority-class improvement, but might sacrifice the overall accuracy. Let the population means of the minority-class true positive rates in the two tasks be $\mu_1$ and $\mu_2$, with $\mu_2 > \mu_1$ typically holding due to the weighting in (4).

When the cross-task genetic transfer is triggered, specifically for individuals in $\mathcal{P}_1$ with minority true positive rate below $\mu_1$, the expected minority-class true positive rates in $\mathcal{P}_1$ after reproduction could be expressed as follows:

$$E[\mu_1'] = \mu_1 + p_{tr}(\mu_2 - \mu_1) \tag{6}$$

where $p_{tr}$ is the transfer probability proportional to the excess minority-class error $(1 - (\mathrm{TP}_k/|S_k|) - \overline{\mathrm{err}}_k)$. Since $\mu_2 > \mu_1$ and $p_{tr} > 0$, it follows that $E[\mu_1'] > \mu_1$, implying an *expected positive drift* in minority-class performance.

Meanwhile, the constrained multiobjective formulation of $\mathcal{T}_1$ ensures that infeasible solutions with inferior overall accuracy are eliminated, thus preventing negative transfer. Therefore, the cross-task genetic transfer mechanism adaptively exchanges complementary information between the two tasks: it injects minority-informative features from $\mathcal{T}_2$ into $\mathcal{T}_1$, correcting imbalance bias, while maintaining overall classification performance through the feasibility constraint. This interaction theoretically explains why the cross-task genetic transfer strategy could enhance both the minority-class and overall classification performance.

### E. Computational Complexity Analysis

Considering one generation, the computational cost of CIL-MTFS encompasses the following parts.

1) *Initialization*: Since tasks $\mathcal{T}_1$ and $\mathcal{T}_2$ share the same initialized population, initialization occurs only once and has a computational complexity of $O(N)$, where $N$ is the population size.
2) *Mating Selection*: For both tasks $\mathcal{T}_1$ and $\mathcal{T}_2$, the computational complexity of selecting mating parents is $O(N)$.
3) *Offspring Generation*: This process happens once per generation with a computational complexity of $O(DN)$, where $D$ is the number of features.
4) *Environmental Selection*: The worst-case computational complexities of environmental selection are $O(N^2)$ and $O(N \log N)$ for tasks $\mathcal{T}_1$ and $\mathcal{T}_2$, respectively.

Note that evaluating the objective values, i.e., training a classification model on each candidate feature subset, is the most computationally expensive component of an evolutionary feature selection algorithm. However, this cost is omitted here since the same classification algorithm is used across all compared methods. Combining all the above components, the overall computational complexity of the proposed CIL-MTFS algorithm per generation is bounded by

TABLE I
PROPERTIES OF THE IMBALANCED CLASSIFICATION DATASETS

| Dataset | #Features | #Samples | Imbalance Ratio |
|---|---|---|---|
| Parkinson | 22 | 195 | 3.06 |
| German | 24 | 1000 | 2.33 |
| WBCD | 30 | 569 | 1.68 |
| Ionosphere | 34 | 351 | 1.79 |
| Lung | 56 | 32 | 2.56 |
| Semeion | 265 | 1593 | 9.08 |
| LSVT | 310 | 126 | 2.00 |
| Armstrong-2002-v1 | 1081 | 72 | 2.00 |
| Gordon-2002 | 1626 | 181 | 4.84 |
| Colon | 2000 | 62 | 1.82 |
| Yeoh-2002-v1 | 2526 | 248 | 4.77 |
| DLBCL | 5469 | 77 | 3.05 |
| CNS | 7129 | 60 | 1.86 |
| Orlraws10P | 10 304 | 100 | 1.00 |
| Brain2 | 10 367 | 50 | 2.14 |
| 11Tumor | 11 340 | 111 | 4.64 |
| GLI-85 | 22 283 | 85 | 2.27 |

$\max\{O(DN), O(N^2)\}$, which aligns with the complexity of its baseline, NSGA-II.

## IV. EXPERIMENT DESIGN

This section provides a detailed description of the datasets used, the baseline methods for comparison, the specific parameter settings, and the performance metrics employed in the experiments.

### A. Datasets

Table I displays 17 real-world imbalanced classification datasets[1,2] from different domains, such as biomedical, finance, radar signal processing, handwritten digit recognition, and face recognition. These datasets encompass a wide range of classification tasks with varying degrees of class imbalance and feature dimensionality. The number of features ranges from 22 in the Parkinson dataset to 22 283 in the GLI-85 dataset, while the number of samples varies from 32 in the Lung dataset to 1593 in the Semeion dataset. The imbalance ratios (IRs) also differ significantly, with Semeion having the highest IR at 9.08 and WBCD the lowest at 1.68. As a result, they can comprehensively test the performance of different compared methods.

### B. Compared Methods

Since the proposed CIL-MTFS method is based on multiobjective optimization, we compare the performance of CIL-MTFS with five multiobjective-based feature selection methods: NSGA-II [47], DAEA [46], MFFS [43], BSOEA [48], and PRDH [49], and also two single-objective-based feature selection methods: IMFWA [35] and MEL [42]. All these methods use the balanced classification error rate in (2) as their first objective, enabling them to handle imbalanced classification tasks. Specifically, MFFS also employs a dual-task approach similar to CIL-MTFS, but its auxiliary task focuses on enhancing the feature subset diversity for the target task. We include NSGA-II as a competitor because the environmental selection of task $\mathcal{T}_1$ in our method is based on NSGA-II. DAEA, BSOEA, PRDH, and IMFWA are included as state-of-the-art feature selection methods, offering a strong comparison for evaluating the effectiveness of CIL-MTFS.

### C. Parameter Settings

During the feature subset search period, each dataset is randomly split into a training data subset (approximately 70%) and a test data subset (approximately 30%), respectively. KNN ($K = 5$) and weighted SVM with five-fold cross-validation are utilized as base classification algorithms to calculate the classification error rate on the training data subset [50]. The reason for using K-fold cross-validation instead of stratified cross-validation is that, stratified cross-validation enforces a fixed class distribution in each fold, which could lead to overfitting to artificially balanced data. In contrast, K-fold cross-validation allows for varying class distributions across folds, better simulating real-world imbalanced scenarios and providing a more robust evaluation of model performance.

Each feature selection method is executed independently 30 times, with the average performance reported. The maximum number of generations for all methods is set to 100. The population size for each algorithm is equal to the number of features ($N = D$), but is capped at 200 ($N = 200$ if $D > 200$) to prevent high computational costs for high-dimensional datasets. Except for IMFWA and MEL, the single-point crossover and bit-flip mutation operators are employed to generate offspring solutions, with the crossover and mutation probabilities are 1.0 and $1/D$, respectively. The nondominated feature subsets obtained by each multiobjective-based method during training are evaluated on the unseen test set. The customizable parameters for each algorithm are set according to the values used in their original articles, which have demonstrated their effectiveness. All experiments are conducted in a Linux environment with an Intel Core i7-11700K, 3.6GHz CPU.

### D. Performance Metrics

The performance of each method is evaluated using the following three commonly used metrics for class-imbalanced learning.

1) *Minimal Balanced Classification Error Rate (MCER)*: Among the multiple nondominated feature subsets from the training set, the feature subset with the lowest training classification error rate (2) is selected, and its testing classification error rate is recorded as MCER.
2) $F_1$*-Score*: It is a widely used metric for imbalanced data, which calculates the harmonic mean of precision and recall. Among the multiple nondominated feature subsets from the training set, the feature subset with the lowest MCER is selected and its testing $F_1$-score is recorded

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{7}$$

[1]https://archive.ics.uci.edu/datasets
[2]http://www.gems-system.org/

where $\text{Precision} = \text{TP}/(\text{TP}+\text{FP})$ and $\text{Recall} = \text{TP}/(\text{TP}+\text{FN})$. Note that (7) is mainly for binary classification tasks. For multiclass scenarios, we adopt the macro $F_1$-score as a replacement.

3) *Geometric Mean (G-mean)*: It is also a commonly used metric for imbalanced data, which calculates the geometric mean of the accuracies for each class. Among the multiple nondominated feature subsets from the training set, the feature subset with the lowest training MCER is selected and its testing $G$-mean is recorded

$$\text{G-mean} = \sqrt{\frac{\text{TP}}{\text{TP}+\text{FN}} \times \frac{\text{TP}}{\text{TN}+\text{FP}}}. \tag{8}$$

Note that (8) is mainly for binary classification tasks. For multiclass scenarios, we adopt the macro $G$-mean as a replacement.

We use Wilcoxon's rank-sum test with a 0.05 significance level to compare the proposed method against each baseline. The symbols "+", "–", and "≈" denote that the proposed method performs significantly better, significantly worse, and is statistically indistinguishable from the baseline, respectively. In addition, Friedman's test is used to sort the performance of all compared methods.

## V. Results Discussions and Analysis

This section presents four experiments designed to address the following research questions.

1) Can the proposed method significantly reduce the number of selected features while maintaining or even improving classification accuracy compared to using all features?
2) How does the performance of the proposed method compare with state-of-the-art methods in terms of classification performance metrics?
3) How do the solution distributions of different methods differ in the objective space?
4) How does the proposed multitask approach compare to single-task approaches?

### A. Comparison With the Full Set

To evaluate whether the proposed CIL-MTFS method can maintain or even improve classification performance while significantly reducing the number of selected features, we first compare its performance with that of the full set (using all original features $\boldsymbol{x}_{\text{full}}$). Table II presents the results of the full set and CIL-MTFS in terms of MCER, $F_1$-score, G-mean, and the number of selected features (abbreviated as FN) metrics.

From the table, it can be seen that the proposed CIL-MTFS method outperforms the full feature set in terms of classification performance (i.e., MCER, $F_1$-score, and G-mean) on at least 12 out of the 17 datasets, while significantly reducing the number of features used. For instance, on the Yeoh-2002-v1 dataset, CIL-MTFS selects only about 51 features out of the original 5469 features, yet achieves a 4.76% decrease in classification error rate. Similarly, on the high-dimensional CNS dataset, CIL-MTFS selects approximately 1053 features from the original 7129 features, with a 3.85% improvement in classification accuracy. However, we also observe that on a few datasets, the

TABLE II
Comparative Results of the Full Set and CIL-MTFS on the Test Data (in Terms of KNN)

| Dataset | Metric | Full Set | CIL-MTFS |
|---|---|---|---|
| Parkinson | MCER | 19.939% | 23.123% |
| | $F_1$-score | 6.9574e–1 | 6.7663e–1 |
| | G-mean | 6.9633e–1 | 6.8716e–1 |
| | FN | 22 | 1.94 |
| German | MCER | 29.397% | 29.016% |
| | $F_1$-score | 5.4900e–1 | 5.5143e–1 |
| | G-mean | 5.5000e–1 | 5.5263e–1 |
| | FN | 24 | 1.70 |
| WBCD | MCER | 5.2984% | 5.1411% |
| | $F_1$-score | 9.1311e–1 | 9.2039e–1 |
| | G-mean | 9.1346e–1 | 9.2116e–1 |
| | FN | 30 | 2.72 |
| Ionosphere | MCER | 33.650% | 7.8876% |
| | $F_1$-score | 6.7734e–1 | 8.6358e–1 |
| | G-mean | 6.8254e–1 | 8.6863e–1 |
| | FN | 34 | 1.72 |
| Lung | MCER | 53.501% | 52.431% |
| | $F_1$-score | 1.8158e–1 | 1.9333e–1 |
| | G-mean | 1.8274e–1 | 1.9732e–1 |
| | FN | 56 | 1.68 |
| Semeion | MCER | 6.7328% | 5.5765% |
| | $F_1$-score | 7.5365e–1 | 8.0744e–1 |
| | G-mean | 7.6348e–1 | 8.1342e–1 |
| | FN | 265 | 25.18 |
| LSVT | MCER | 25.768% | 18.952% |
| | $F_1$-score | 6.8975e–1 | 7.1309e–1 |
| | G-mean | 6.9013e–1 | 7.1999e–1 |
| | FN | 310 | 92.29 |
| Armstrong-2002-v1 | MCER | 14.298% | 20.440% |
| | $F_1$-score | 8.3333e–1 | 7.3569e–1 |
| | G-mean | 8.4523e–1 | 7.4928e–1 |
| | FN | 1081 | 156.42 |
| Gordon-2002 | MCER | 2.0899% | 4.7882% |
| | $F_1$-score | 8.5611e–1 | 8.5500e–1 |
| | G-mean | 8.5603e–1 | 8.6210e–1 |
| | FN | 1626 | 349.10 |
| Colon | MCER | 33.853% | 25.032% |
| | $F_1$-score | 6.0562e–1 | 6.3480e–1 |
| | G-mean | 6.0892e–1 | 6.5521e–1 |
| | FN | 2000 | 758.60 |
| Yeoh-2002-v1 | MCER | 7.5376% | 2.7779% |
| | $F_1$-score | 9.5384e–1 | 9.7063e–1 |
| | G-mean | 9.5887e–1 | 9.7118e–1 |
| | FN | 2526 | 51.28 |
| DLBCL | MCER | 22.277% | 23.026% |
| | $F_1$-score | 6.6667e–1 | 5.4961e–1 |
| | G-mean | 7.0711e–1 | 5.7501e–1 |
| | FN | 5469 | 835.12 |
| CNS | MCER | 14.290% | 10.437% |
| | $F_1$-score | 7.9441e–1 | 8.1747e–1 |
| | G-mean | 8.0102e–1 | 8.2493e–1 |
| | FN | 7129 | 1053.07 |
| Orlraws10P | MCER | 11.670% | 18.616% |
| | $F_1$-score | 9.5761e–1 | 8.7016e–1 |
| | G-mean | 9.6591e–1 | 8.4317e–1 |
| | FN | 10304 | 29.60 |
| Brain2 | MCER | 35.566% | 34.221% |
| | $F_1$-score | 5.0504e–1 | 5.7804e–1 |
| | G-mean | 7.2931e–1 | 6.0026e–1 |
| | FN | 10367 | 68.50 |
| 11Tumor | MCER | 35.263% | 30.968% |
| | $F_1$-score | 5.9239e–1 | 6.2613e–1 |
| | G-mean | 3.2405e-2 | 9.2860e-2 |
| | FN | 12533 | 281.43 |
| GLI-85 | MCER | 24.569% | 18.895% |
| | $F_1$-score | 6.1548e–1 | 6.9104e–1 |
| | G-mean | 6.1725e–1 | 6.9731e–1 |
| | FN | 22283 | 12463.93 |

Note: The best result is highlighted with a gray background.

error rate increases after feature selection, despite a substantial reduction in the number of features selected. Therefore, placing greater emphasis on classification performance rather than solely on feature reduction remains an important direction for our future work.

On the WBCD dataset, the task is to classify breast cancer cases as malignant or benign based on cell features. Although the classification error rate after feature selection (5.1411%) is comparable to that using all features (5.2984%), CIL-MTFS selects only an average of 2.72 features out of the original 30. This not only achieves a substantial reduction in feature dimensionality, but also provides valuable insight by pinpointing the few cell features most indicative of a malignant or benign diagnosis.

The above results demonstrate that the proposed CIL-MTFS method effectively identifies a small subset of valuable features from the original full set, while even improving classification performance compared to using all features.

### B. Comparison in Terms of the Classification Performance

To verify the performance of CIL-MTFS compared to state-of-the-art multiobjective-based feature selection methods, Table III presents the comparative average $F_1$-scores achieved by NSGA-II, DAEA, MFFS, BSOEA, PRDH, IMFWA, and CIL-MTFS on the test data, with the best results for each dataset highlighted in bold. As shown in Table III, among the seven compared state-of-the-art methods, the proposed CIL-MTFS method can obtain the best results on 7 out of the 17 datasets. The Wilcoxon's rank-sum test results indicate that CIL-MTFS significantly outperforms NSGA-II, DAEA, MFFS, BSOEA, PRDH, and IMFWA on 8, 5, 4, 7, 5, and 7 datasets, respectively. In contrast, no competing algorithm shows significantly better performance than CIL-MTFS on more than two datasets. Furthermore, the results of the Friedman's test at the bottom of Table III confirm that CIL-MTFS ranks first among the seven algorithms, further highlighting its superior performance in terms of the $F_1$-score metric.

Similar patterns are observed in terms of the G-mean and MCER metrics, as shown in Tables IV and V, respectively. A closer examination of these results reveals that the performance of CIL-MTFS is particularly notable on high-dimensional datasets, such as Yeoh-2002-v1, DLBCL, CNS, and GLI-85, which contain 2526, 5469, 7129, and 22 283 features, respectively. These datasets also have far fewer samples than features, resulting in highly sparse data spaces that present significant challenges for various feature selection methods. Despite this, CIL-MTFS achieves the best overall classification performance on these datasets, demonstrating its effectiveness in handling high-dimensional imbalanced data.

### C. Analysis of Nondominated Feature Subsets on the Test Set

As a multiobjective feature selection method, it is essential to analyze the distribution of nondominated solutions (e.g., feature subsets) within the objective space. To assess the distribution of feature subsets, Fig. 2 illustrates the nondominated feature subsets obtained by NSGA-II, DAEA, MFFS, BSOEA, PRDH, and CIL-MTFS on the WBCD, Semeion, CNS, and GLI-85 datasets (on the test set). These datasets represent small, medium, and high-dimensional data. Note that the patterns observed for these algorithms on other datasets are similar. In Fig. 2, the black solid line denotes the classification error rate when using all features ($\boldsymbol{x}_{\text{full}}$). It is also important to clarify that the nondominated feature subsets shown for the test set are derived from a two-step process. First, we identify the nondominated solutions on the training set, denoted as $\mathcal{S}_{\text{train}}$. These solutions are then evaluated on the unseen test set. During this evaluation, some solutions in $\mathcal{S}_{\text{train}}$ might become dominated, resulting in a reduced number of nondominated solutions on the test set. For clearer visualization, we exclude the dominated ones and display only the remaining nondominated solutions.

From Fig. 2, it is evident that the nondominated feature subsets obtained by the proposed CIL-MTFS method consistently exhibit balanced classification error rates that are smaller than or comparable to those achieved using all features. In contrast, some of the feature subsets obtained from other methods show extremely higher error rates than when using all features, primarily because these methods do not incorporate constraints and treat both objectives equally. For example, on the Semeion dataset, the classification error rate using all features is approximately 6.7328%. The nondominated feature subsets from CIL-MTFS are all below this threshold, whereas most feature subsets from the other methods exceed 6.7328%, with some even surpassing 50%. Although these feature subsets are nondominated, they are clearly impractical in practice. The goal of feature selection is to eliminate redundant and irrelevant features while maintaining or even improving classification accuracy. The results in Fig. 2 demonstrate that the proposed method successfully achieves this goal, offering a set of *meaningful* feature subsets: those with error rates equal to or lower than that of using all features. This not only helps decision makers by providing more useful feature subsets, but also reduces the selection burden by eliminating meaningless nondominated feature subsets.

From Fig. 2, we also notice that the feature subsets selected by the proposed CIL-MTFS method tend to include more features, which is one limitation of CIL-MTFS. This can be attributed to the constraint imposed on task $\mathcal{T}_1$, where feature subsets that select fewer features but violate the constraint are more likely to be discarded during the selection process. As a result, the population contains more feasible solutions (i.e., feature subsets with error rates lower than that of using all features, but tends to select a larger number of features). However, of the two major objectives in multiobjective feature selection, classification performance (i.e., classification error rate) holds greater importance than the number of selected features. Therefore, sacrificing an increase in the number of selected features to improve classification accuracy is a justifiable tradeoff.

Another limitation of the proposed CIL-MTFS method is its focus on two-class tasks, as shown in Table I. For tasks involving multiple classes or multiple minority classes, one potential solution is to group samples into majority and minority classes. The proposed framework could then be adapted to address such scenarios. This extension will be explored in future work.

TABLE III
COMPARATIVE RESULTS OF AVERAGE $F_1$-SCORE VALUES OBTAINED BY NSGA-II, DAEA, MFFS, BSOEA, PRDH, IMFWA, AND CIL-MTFS ON THE TEST DATA (IN TERMS OF KNN)

| Data set | NSGA-II [47] | DAEA [46] | MFFS [43] | BSOEA [48] | PRDH [49] | IMFWA [35] | CIL-MTFS |
|---|---|---|---|---|---|---|---|
| Parkinson | 6.2177e–1+ | 6.6362e–1+ | 6.2430e–1+ | 6.7591e–1≈ | 6.5791e–1≈ | 6.9920e–1– | 6.7663e–1 |
| German | 5.3655e–1+ | 5.5132e–1≈ | 5.5077e–1≈ | 5.4962e–1≈ | 5.5263e–1≈ | 5.1663e–1+ | 5.5143e–1 |
| WBCD | 9.2641e–1– | 9.1655e–1≈ | 9.2111e–1≈ | 9.2053e–1≈ | 9.2011e–1≈ | 9.2180e–1≈ | 9.2039e–1 |
| Ionosphere | 8.5126e–1+ | 8.5864e–1+ | 8.5577e–1+ | 8.5144e–1+ | 8.5969e–1+ | 8.3359e–1+ | 8.6358e–1 |
| Lung | 4.0000e-2+ | 1.2000e–1+ | 1.7333e–1≈ | 9.3333e-2+ | 1.4667e–1+ | 3.9838e-2+ | 1.9333e–1 |
| Semeion | 8.6973e–1≈ | 8.1348e–1≈ | 8.6796e–1≈ | 8.0070e–1≈ | 8.8370e–1≈ | 7.9059e–1≈ | 8.0744e–1 |
| LSVT | 7.1317e–1≈ | 6.9578e–1+ | 7.1835e–1≈ | 6.8217e–1+ | 7.3161e–1≈ | 6.8105e–1+ | 7.1309e–1 |
| Armstrong-2002-v1 | 7.4336e–1≈ | 7.5684e–1≈ | 7.5865e–1≈ | 7.3773e–1≈ | 7.0610e–1≈ | 7.9933e–1– | 7.3569e–1 |
| Gordon-2002 | 8.7404e–1≈ | 8.5922e–1≈ | 8.2309e–1≈ | 8.6251e–1≈ | 8.6929e–1≈ | 8.8978e–1≈ | 8.5500e–1 |
| Colon | 5.9591e–1≈ | 6.0896e–1≈ | 5.8030e–1≈ | 6.8099e–1≈ | 6.2598e–1≈ | 5.3444e–1+ | 6.3480e–1 |
| Yeoh-2002-v1 | 9.6910e–1≈ | 9.6694e–1≈ | 9.7028e–1≈ | 9.5902e–1+ | 9.6584e–1≈ | 9.4967e–1+ | 9.7063e–1 |
| DLBCL | 4.5146e–1+ | 5.2437e–1≈ | 5.1908e–1≈ | 5.0721e–1≈ | 5.4034e–1≈ | 5.0565e–1≈ | 5.4961e–1 |
| CNS | 3.8589e–1+ | 8.1430e–1≈ | 7.8177e–1≈ | 4.2635e–1+ | 8.1135e–1≈ | 8.1155e–1≈ | 8.1747e–1 |
| Orlraws10P | 9.3479e–1– | 7.8673e–1+ | 8.2828e–1≈ | 8.2681e–1+ | 7.6252e–1+ | 8.8180e–1≈ | 8.7016e–1 |
| Brain2 | 5.2331e–1+ | 5.2727e–1≈ | 5.9087e–1≈ | 5.6462e–1≈ | 5.3720e–1≈ | 5.4345e–1≈ | 5.7804e–1 |
| 11Tumor | 5.9524e–1+ | 6.0594e–1≈ | 5.9141e–1+ | 5.9562e–1+ | 5.8656e–1+ | 5.5876e–1+ | 6.2613e–1 |
| GLI-85 | 6.7204e–1≈ | 6.8145e–1≈ | 6.4504e–1+ | 6.7187e–1≈ | 6.4577e–1+ | 6.8390e–1≈ | 6.9104e–1 |
| +/ – / ≈ | 8/2/7 | 5/0/11 | 4/0/12 | 7/0/10 | 5/0/12 | 7/2/8 | – |
| **Friedman's rank** | 4.4118 | 4.0000 | 3.9412 | 4.5294 | 3.9412 | 4.5882 | 2.5882 |

Note: The best result is highlighted with a gray background.

TABLE IV
COMPARATIVE RESULTS OF AVERAGE G-MEAN VALUES OBTAINED BY NSGA-II, DAEA, MFFS, BSOEA, PRDH, IMFWA, AND CIL-MTFS ON THE TEST DATA (IN TERMS OF KNN)

| Data set | NSGA-II [47] | DAEA [46] | MFFS [43] | BSOEA [48] | PRDH [49] | IMFWA [35] | CIL-MTFS |
|---|---|---|---|---|---|---|---|
| Parkinson | 6.2415e–1+ | 6.7199e–1+ | 6.2721e–1+ | 6.8696e–1≈ | 6.6451e–1≈ | 7.0431e–1– | 6.8716e–1 |
| German | 5.3787e–1+ | 5.5268e–1≈ | 5.5196e–1≈ | 5.5100e–1≈ | 5.5383e–1≈ | 5.1667e–1+ | 5.5263e–1 |
| WBCD | 9.2682e–1– | 9.1743e–1≈ | 9.2172e–1≈ | 9.2174e–1≈ | 9.2067e–1≈ | 9.2273e–1≈ | 9.2116e–1 |
| Ionosphere | 8.5642e–1+ | 8.6368e–1+ | 8.6092e–1+ | 8.5684e–1+ | 8.6463e–1+ | 8.5424e–1+ | 8.6863e–1 |
| Lung | 4.2159e-2+ | 1.2247e–1+ | 1.7691e–1≈ | 9.5258e-2+ | 1.4969e–1+ | 4.2016e-2+ | 1.9732e–1 |
| Semeion | 8.7444e–1≈ | 8.1920e–1≈ | 8.7206e–1≈ | 8.0746e–1≈ | 8.8707e–1≈ | 8.0175e–1≈ | 8.1342e–1 |
| LSVT | 7.2042e–1≈ | 7.0238e–1≈ | 7.2549e–1≈ | 6.8961e–1+ | 7.4060e–1≈ | 6.8105e–1+ | 7.1999e–1 |
| Armstrong-2002-v1 | 7.6325e–1≈ | 7.7731e–1≈ | 7.7859e–1≈ | 7.5506e–1≈ | 7.1923e–1≈ | 7.9901e–1– | 7.4928e–1 |
| Gordon-2002 | 8.8201e–1≈ | 8.6445e–1≈ | 8.2918e–1≈ | 8.6729e–1≈ | 8.7573e–1≈ | 8.8990e–1≈ | 8.6210e–1 |
| Colon | 6.2521e–1≈ | 6.3800e–1≈ | 6.0749e–1≈ | 7.0008e–1≈ | 6.5618e–1≈ | 5.3458e–1+ | 6.5521e–1 |
| Yeoh-2002-v1 | 9.6963e–1≈ | 9.6753e–1≈ | 9.7104e–1≈ | 9.5954e–1+ | 9.6641e–1≈ | 9.5748e–1+ | 9.7118e–1 |
| DLBCL | 4.7369e–1+ | 5.5648e–1≈ | 5.4921e–1≈ | 5.3124e–1≈ | 5.6865e–1≈ | 5.0692e–1≈ | 5.7501e–1 |
| CNS | 3.9820e–1+ | 8.1748e–1≈ | 7.8893e–1≈ | 4.3361e–1+ | 8.1735e–1≈ | 8.2081e–1≈ | 8.2493e–1 |
| Orlraws10P | 9.5277e–1– | 6.4923e–1+ | 7.6308e–1≈ | 6.9473e–1+ | 5.6298e–1+ | 8.6521e–1≈ | 8.4317e–1 |
| Brain2 | 7.3802e–1≈ | 5.1773e–1≈ | 5.8768e–1≈ | 5.6428e–1≈ | 5.0394e–1≈ | 6.6687e–1≈ | 6.0026e–1 |
| 11Tumor | 7.6630e-2≈ | 7.7509e-2≈ | 6.3575e-2+ | 7.0645e–1≈ | 7.0415e-2+ | 7.0324e-2+ | 9.2860e-2 |
| GLI-85 | 6.8063e–1≈ | 7.0715e–1≈ | 6.5208e–1+ | 6.8107e–1≈ | 6.5337e–1+ | 6.8902e–1≈ | 6.9731e–1 |
| +/ – / ≈ | 6/2/9 | 4/0/13 | 4/0/13 | 6/0/11 | 5/0/12 | 7/2/8 | – |
| **Friedman's rank** | 4.1176 | 3.9412 | 4.2941 | 4.4118 | 3.9412 | 4.4118 | 2.8824 |

Note: The best result is highlighted with a gray background.

### D. Analysis Based on the Multiobjective Indicator

As a multiobjective-based feature selection method, CIL-MTFS should also be evaluated using multiobjective performance indicators. The comparative results based on the HV metric are presented in Tables S-VIII and S-IX of the supplement document. Overall, the Friedman's test results show that CIL-MTFS ranks third and fifth when using KNN and weighted SVM classifiers, respectively. The relatively modest HV performance is expected, as our method differs fundamentally from conventional multiobjective approaches that aim to explore the entire Pareto front. Specifically, CIL-MTFS formulates feature selection as a constrained multiobjective optimization problem, introducing a constraint to eliminate feature subsets that select very few features yet perform worse than using all features. Although such subsets could be nondominated in the traditional sense, they are practically meaningless and thus excluded. As a result, the algorithm focuses on a smaller but more meaningful region of the objective space, emphasizing solutions that outperform the full feature set. This focused search naturally yields worse HV values, since only a subset of the Pareto front is explored. As illustrated in Fig. 2, on the Semeion dataset, CIL-MTFS identifies feature subsets with comparable or even lower error rates than those obtained using all features, whereas some other algorithms explore the entire Pareto front and generate many solutions with error rates exceeding 50% (compared to 6.73% using all features), which offer limited practical value.

Fig. 3 presents the convergence profiles of CIL-MTFS in terms of the HV metric. The figure illustrates the evolution of

TABLE V
COMPARATIVE RESULTS OF AVERAGE MCER VALUES OBTAINED BY NSGA-II, DAEA, MFFS, BSOEA, PRDH, IMFWA, AND CIL-MTFS ON THE TEST DATA (IN TERMS OF KNN)

| **Dataset** | **NSGA-II** [47] | **DAEA** [46] | **MFFS** [43] | **BSOEA** [48] | **PRDH** [49] | **IMFWA** [35] | **CIL-MTFS** |
|---|---|---|---|---|---|---|---|
| Parkinson | 24.818%+ | 23.798%+ | 24.809%+ | 23.309%≈ | 23.751%≈ | 20.123%− | 23.123% |
| German | 30.056%≈ | 28.941%≈ | 29.075%≈ | 29.036%≈ | 28.938%≈ | 31.647%+ | 29.016% |
| WBCD | 5.3126%+ | 5.4339%≈ | 5.4490%≈ | 4.8541%≈ | 5.5451%≈ | 4.8877%≈ | 5.1411% |
| Ionosphere | 8.8623%+ | 8.2969%+ | 8.4832%+ | 8.7466%+ | 8.2542%+ | 9.5587%+ | 7.8876% |
| Lung | 59.086%+ | 56.250%+ | 53.472%≈ | 57.639%+ | 54.861%+ | 59.454%+ | 52.431% |
| Semeion | 3.9202%≈ | 5.2923%≈ | 3.9478%≈ | 5.4851%≈ | 3.7073%− | 5.8231%≈ | 5.5765% |
| LSVT | 18.965%≈ | 20.330%≈ | 18.649%≈ | 20.988%≈ | 16.724%≈ | 20.341%≈ | 18.952% |
| Armstrong-2002-v1 | 20.376%≈ | 19.524%≈ | 19.371%≈ | 20.621%≈ | 22.387%≈ | 14.375%− | 20.440% |
| Gordon-2002 | 3.4176%≈ | 5.6510%≈ | 7.0746%+ | 5.3712%≈ | 4.6279%≈ | 3.5591%≈ | 4.7882% |
| Colon | 27.659%≈ | 26.599%≈ | 29.090%≈ | 21.508%≈ | 26.239%≈ | 29.500%+ | 25.032% |
| Yeoh-2002-v1 | 2.7473%≈ | 2.5122%≈ | 2.8098%≈ | 2.8059%≈ | 3.0716%≈ | 3.7351%≈ | 2.7779% |
| DLBCL | 31.668%+ | 27.237%+ | 23.534%≈ | 26.954%≈ | 23.077%≈ | 26.544%≈ | 23.026% |
| CNS | 40.917%+ | 12.230%≈ | 12.914%≈ | 40.221%+ | 11.573%≈ | 10.726%≈ | 10.437% |
| Orlraws10P | 13.222%− | 26.083%+ | 22.015%≈ | 22.820%≈ | 27.822%+ | 17.884%≈ | 18.616% |
| Brain2 | 29.717%≈ | 42.293%+ | 36.147%≈ | 35.379%≈ | 41.361%+ | 31.674%≈ | 34.221% |
| 11Tumor | 32.724%≈ | 34.284%+ | 35.527%+ | 34.581%+ | 36.772%+ | 36.087%+ | 30.968% |
| GLI-85 | 19.518%≈ | 19.489%≈ | 23.635%+ | 19.789%≈ | 22.964%+ | 19.408%≈ | 18.895% |
| +/ − / ≈ | 5/1/11 | 7/0/10 | 5/0/12 | 4/0/13 | 6/1/10 | 5/2/10 | – |
| **Friedman's rank** | 4.0588 | 4.1765 | 4.5294 | 4.4118 | 4.0588 | 4.2353 | 2.5294 |

Note: The best result is highlighted with a gray background.

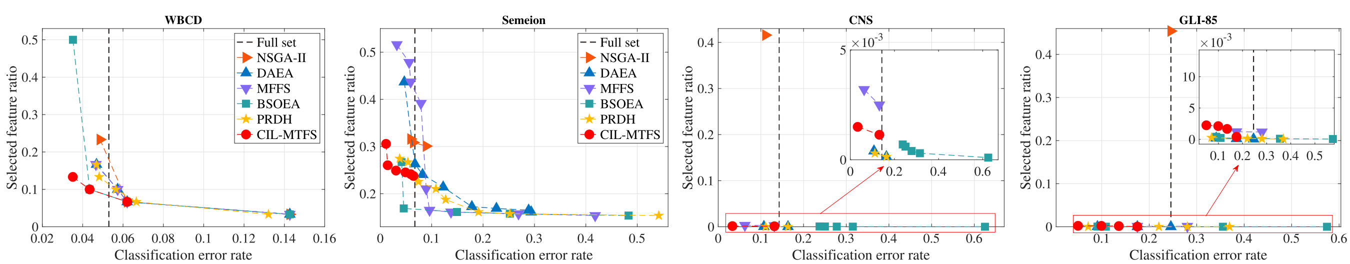


Fig. 2. Nondominated feature subsets on the test set, where the black dashed line represents the classification error rate using all features.

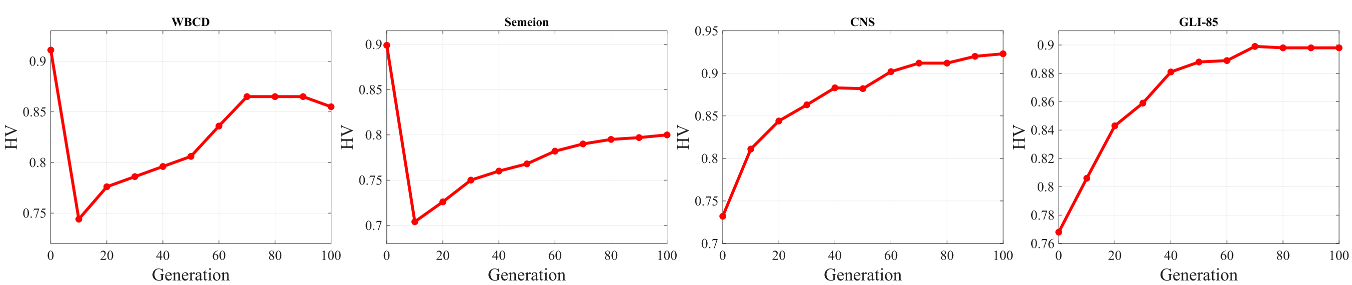


Fig. 3. Convergence curve of CIL-MTFS in terms of HV on the training data.

HV values over generations during the training process on four representative datasets: WBCD, Semeion, CNS, and GLI-85, which correspond to small-, medium-, and high-dimensional data, respectively. An interesting pattern can be observed on the WBCD and Semeion datasets: the HV values initially decrease and then increase steadily as evolution progresses. This behavior arises because, in the early generations, the nondominated solutions identified by CIL-MTFS often include infeasible solutions, i.e., feature subsets with error rates higher than those obtained using all features. Although these infeasible solutions can yield higher HV values, they are progressively removed as the proposed CIL-MTFS method prioritizes feasible and practically meaningful solutions. Consequently, the initial decline in HV reflects this pruning process, followed by a steady increase as the population converges toward high-quality, feasible feature subsets.

### E. Ablation Study

To evaluate the effectiveness of the proposed multitask approach, this subsection compares the performance of CIL-MTFS with that of single-task approaches. Since the constraint imposed in $\mathcal{T}_1$ has proven effective in filtering undesirable feature subsets for decision makers, we retain this constraint in our experiments. Specifically, the first single-task algorithm uses only the constrained multiobjective feature selection formulation from $\mathcal{T}_1$, referred to as CIL-T$_1$FS. In

TABLE VI
COMPARATIVE RESULTS OF CIL-T$_1$FS, CIL-T$_2$FS, AND CIL-MTFS ON THE TEST DATA (IN TERMS OF KNN)

| Dataset | Metric | CIL-T$_1$FS | CIL-T$_2$FS | CIL-MTFS |
|---|---|---|---|---|
| Parkinson | MCER | 23.886%+ | 23.232%≈ | 23.123% |
| | $F_1$-score | 6.6469e−1+ | 6.5764e−1≈ | 6.7663e−1 |
| | G-mean | 6.7387e−1+ | 6.6286e−1≈ | 6.8716e−1 |
| German | MCER | 30.089%≈ | 30.067%≈ | 29.016% |
| | $F_1$-score | 5.3551e−1+ | 5.3514e−1+ | 5.5143e−1 |
| | G-mean | 5.3694e−1+ | 5.3652e−1+ | 5.5263e−1 |
| WBCD | MCER | 5.2704%≈ | 4.9081%≈ | 5.1411% |
| | $F_1$-score | 9.2387e−1≈ | 9.2511e−1≈ | 9.2039e−1 |
| | G-mean | 9.2443e−1≈ | 9.2576e−1≈ | 9.2116e−1 |
| Ionosphere | MCER | 8.7049%+ | 7.1912%− | 7.8876% |
| | $F_1$-score | 8.4930e−1+ | 8.6708e−1≈ | 8.6358e−1 |
| | G-mean | 8.5520e−1+ | 8.7326e−1≈ | 8.6863e−1 |
| Lung | MCER | 56.809%+ | 57.156%+ | 52.431% |
| | $F_1$-score | 8.8889e-2+ | 8.4815e-2+ | 1.9333e−1 |
| | G-mean | 9.1612e-2+ | 8.6813e-2+ | 1.9732e−1 |
| Semeion | MCER | 6.6586%+ | 5.8649%≈ | 5.5765% |
| | $F_1$-score | 8.0106e−1≈ | 8.0405e−1≈ | 8.0744e−1 |
| | G-mean | 8.0501e−1+ | 8.1143e−1≈ | 8.1342e−1 |
| LSVT | MCER | 19.595%≈ | 21.954%≈ | 18.952% |
| | $F_1$-score | 7.1222e−1≈ | 6.6838e−1+ | 7.1309e−1 |
| | G-mean | 7.1856e−1≈ | 6.7676e−1+ | 7.1999e−1 |
| Armstrong-2002-v1 | MCER | 20.916%≈ | 20.772%≈ | 20.440% |
| | $F_1$-score | 7.2709e−1≈ | 7.3125e−1≈ | 7.3569e−1 |
| | G-mean | 7.3989e−1≈ | 7.4677e−1≈ | 7.4928e−1 |
| Gordon-2002 | MCER | 3.3003%≈ | 3.3696%≈ | 4.7882% |
| | $F_1$-score | 8.8279e−1≈ | 8.7276e−1≈ | 8.5500e−1 |
| | G-mean | 8.8930e−1≈ | 8.7991e−1≈ | 8.6210e−1 |
| Colon | MCER | 25.806%≈ | 24.658%≈ | 25.032% |
| | $F_1$-score | 6.2776e−1≈ | 6.5250e−1≈ | 6.3480e−1 |
| | G-mean | 6.5727e−1≈ | 6.8386e−1≈ | 6.5521e−1 |
| Yeoh-2002-v1 | MCER | 2.8808%≈ | 3.1433%+ | 2.7779% |
| | $F_1$-score | 9.6127e−1+ | 9.6319e−1+ | 9.7063e−1 |
| | G-mean | 9.6213e−1+ | 9.6385e−1+ | 9.7118e−1 |
| DLBCL | MCER | 20.781%≈ | 27.646%≈ | 23.026% |
| | $F_1$-score | 5.5203e−1≈ | 4.9310e−1≈ | 5.4961e−1 |
| | G-mean | 5.8189e−1≈ | 5.1421e−1≈ | 5.7501e−1 |
| CNS | MCER | 41.762%+ | 46.508%+ | 10.437% |
| | $F_1$-score | 4.0530e−1+ | 3.0444e−1+ | 8.1747e−1 |
| | G-mean | 4.1207e−1+ | 3.1482e−1+ | 8.2493e−1 |
| Orlraws10P | MCER | 17.490%≈ | 16.676%≈ | 18.616% |
| | $F_1$-score | 8.8835e−1≈ | 8.9528e−1≈ | 8.7016e−1 |
| | G-mean | 8.3211e−1≈ | 8.6261e−1≈ | 8.4317e−1 |
| Brain2 | MCER | 38.811%≈ | 34.624%≈ | 34.221% |
| | $F_1$-score | 5.4837e−1≈ | 6.1559e−1≈ | 5.7804e−1 |
| | G-mean | 4.0325e−1≈ | 5.2007e−1≈ | 6.0026e−1 |
| 11Tumor | MCER | 35.014%≈ | 33.449%≈ | 30.968% |
| | $F_1$-score | 5.8799e−1+ | 6.0583e−1≈ | 6.2613e−1 |
| | G-mean | 7.3849e−1≈ | 7.7612e−1≈ | 9.2860e-2 |
| GLI-85 | MCER | 23.793%+ | 21.104%+ | 18.895% |
| | $F_1$-score | 6.1850e−1+ | 6.3296e−1+ | 6.9104e−1 |
| | G-mean | 6.2533e−1+ | 6.4662e−1+ | 6.9731e−1 |
| $+/-/\approx$ | MCER | 6/0/11 | 4/1/12 | – |
| | $F_1$-score | 8/0/9 | 6/0/11 | – |
| | G-mean | 8/0/9 | 6/0/11 | – |

Note: The best result is highlighted with a gray background.

the second single-task algorithm, the first objective function in the constrained multiobjective feature selection problem is replaced with the cost-sensitive objective function [i.e., (4)], resulting in the CIL-T$_2$FS algorithm.

Table VI presents the detailed results in terms of MCER, $F_1$-score, and G-mean metrics. The table reveals that, in terms of the MCER metric, the proposed multitask method significantly outperforms the single-task methods, CIL-T$_1$FS and CIL-T$_2$FS, on 6 and 4 out of the 17 datasets, respectively. Regarding the $F_1$-score, the proposed method significantly surpasses CIL-T$_1$FS and CIL-T$_2$FS on 8 and 6 datasets, respectively. Similarly, for the G-mean metric, the proposed CIL-MTFS method demonstrates a clear advantage over CIL-T$_1$FS and CIL-T$_2$FS on 8 and 6 datasets, respectively. It is worth noting that, in contrast, the performance of CIL-MTFS is only significantly worse than that of the single-task methods in one instance: on the Ionosphere dataset, where it performs worse than CIL-T$_2$FS according to the MCER metric.

Based on the above results, among the two single-task methods, CIL-T$_2$FS, which utilizes the cost-sensitive error rate in (4), appears to achieve better classification performance compared to CIL-T$_1$FS, which employs the balanced error rate in (2). This difference in performance could be attributed to the fact that the cost-sensitive error rate used in CIL-T$_2$FS considers the inverse class distribution, giving more emphasis to the performance of the minority class(es). In contrast, the balanced error rate employed in CIL-T$_1$FS is based on an equally weighted class distribution, prioritizing overall class performance. As a result, CIL-T$_2$FS performs better on these imbalanced datasets. However, both CIL-T$_1$FS and CIL-T$_2$FS exhibit worse performance than CIL-MTFS. This is because CIL-MTFS not only considers overall class performance, but also leverages features that are particularly beneficial for the minority class, resulting in improved classification performance for class-imbalanced tasks.

## VI. CONCLUSION

The goal of this article was to address the data challenges posed by class imbalance and high dimensionality. This goal has been successfully achieved by proposing the CIL-MTFS method, which addresses a constrained multiobjective feature selection task and its reconstructed cost-sensitive feature selection task within a multitask setting. Through carefully designed task formulations, each task can focus on different aspects of problem solving. In addition, a knowledge-sharing mechanism is introduced to facilitate the transfer of informative and generalizable features between tasks, enabling the effective reuse of feature subsets and enhancing both overall classification performance and minority class performance. Experimental results on 17 real-world datasets demonstrate that the proposed CIL-MTFS method outperforms the multiobjective-based feature selection methods, achieving the best overall classification performance. Notably, one salient advantage of the proposed method is that the feature subsets it generates can all achieve classification error rates that are either similar to or even lower than that obtained using all features, while significantly reducing data dimensionality.

The work of this article offers several important contributions to the field of evolutionary data mining. First, it introduces a new method for dimensionality reduction in tabular datasets, providing more efficient handling of high-dimensional data. In addition, it promotes multiobjective data mining by presenting decision-makers with a concise set of feature subsets, simplifying the decision-making process. Finally, it deepens the understanding of feature selection and class-imbalanced learning, exploring how their interplay can enhance classification performance.

Future work can be explored from several perspectives. One potential extension is applying the proposed method to long-tailed classification tasks, where a few classes dominate the data distribution, while the majority of classes have significantly fewer samples, resulting in an imbalanced and skewed distribution. In addition, further reducing the number of selected features is another direction for future research. Last, the proposed method can be integrated with sample cleaning and sample selection techniques to address label-noise and class-imbalance tasks [51].

## REFERENCES


[1] H. He and E. A. Garcia, "Learning from imbalanced data," *IEEE Trans. Knowl. Data Eng.*, vol. 21, no. 9, pp. 1263–1284, Sep. 2009.

[2] S. Das, S. S. Mullick, and I. Zelinka, "On supervised class-imbalanced learning: An updated perspective and some key challenges," *IEEE Trans. Artif. Intell.*, vol. 3, no. 6, pp. 973–993, Dec. 2022.

[3] A. Khurana and O. P. Verma, "Optimal feature selection for imbalanced text classification," *IEEE Trans. Artif. Intell.*, vol. 4, no. 1, pp. 135–147, Feb. 2023.

[4] W. Pei, B. Xue, L. Shang, and M. Zhang, "High-dimensional unbalanced binary classification by genetic programming with multi-criterion fitness evaluation and selection," *Evol. Comput. (MIT Press)*, vol. 30, no. 1, pp. 99–129, 2022.

[5] J. Li and H. Liu, "Challenges of feature selection for big data analytics," *IEEE Intell. Syst.*, vol. 32, no. 2, pp. 9–15, Mar./Apr. 2017.

[6] R. Jiao, B. H. Nguyen, B. Xue, and M. Zhang, "A survey on evolutionary multiobjective feature selection in classification: Approaches, applications, and challenges," *IEEE Trans. Evol. Comput.*, vol. 28, no. 4, pp. 1156–1176, Aug. 2024.

[7] V. Borisov, T. Leemann, K. Seßler, J. Haug, M. Pawelczyk, and G. Kasneci, "Deep neural networks and tabular data: A survey," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 6, pp. 7499–7519, Jun. 2024.

[8] Y. Mei, Q. Chen, A. Lensen, B. Xue, and M. Zhang, "Explainable artificial intelligence by genetic programming: A survey," *IEEE Trans. Evol. Comput.*, vol. 27, no. 3, pp. 621–641, Jun. 2023.

[9] B. Xue, M. Zhang, W. N. Browne, and X. Yao, "A survey on evolutionary computation approaches to feature selection," *IEEE Trans. Evol. Comput.*, vol. 20, no. 4, pp. 606–626, Aug. 2016.

[10] A. Telikani, A. Tahmassebi, W. Banzhaf, and A. H. Gandomi, "Evolutionary machine learning: A survey," *ACM Comput. Surv.*, vol. 54, no. 8, pp. 1–35, 2021.

[11] A. E. Eiben and J. Smith, "From evolutionary computation to the evolution of things," *Nature*, vol. 521, no. 7553, pp. 476–482, 2015.

[12] H. Xu, B. Xue, and M. Zhang, "Probe population based initialization and genetic pool based reproduction for evolutionary bi-objective feature selection," *IEEE Trans. Evol. Comput.*, vol. 29, no. 5, pp. 1849–1863, Oct. 2025, doi: 10.1109/TEVC.2024.3403655.

[13] W. Pei, B. Xue, M. Zhang, L. Shang, X. Yao, and Q. Zhang, "A survey on unbalanced classification: How can evolutionary computation help?" *IEEE Trans. Evol. Comput.*, vol. 28, no. 2, pp. 353–373, Apr. 2024.

[14] H. Xu, B. Xue, and M. Zhang, "A bi-search evolutionary algorithm for high-dimensional bi-objective feature selection," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 8, no. 5, pp. 3489–3502,Oct. 2024.

[15] K. Chen, B. Xue, M. Zhang, and F. Zhou, "Evolutionary multitasking for feature selection in high-dimensional classification via particle swarm optimization," *IEEE Trans. Evol. Comput.*, vol. 26, no. 3, pp. 446–460, Jun. 2022.

[16] L. Li, M. Xuan, Q. Lin, M. Jiang, Z. Ming, and K. C. Tan, "An evolutionary multitasking algorithm with multiple filtering for high-dimensional feature selection," *IEEE Trans. Evol. Comput.*, vol. 27, no. 4, pp. 802–816, Aug. 2023.

[17] Y. Sun, K. Tang, L. L. Minku, S. Wang, and X. Yao, "Online ensemble learning of data streams with gradually evolved classes," *IEEE Trans. Knowl. Data Eng.*, vol. 28, no. 6, pp. 1532–1545, Jun. 2016.

[18] Y. Yan, Y. Zhu, R. Liu, Y. Zhang, Y. Zhang, and L. Zhang, "Spatial distribution-based imbalanced undersampling," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 6, pp. 6376–6391, Jun. 2023.

[19] T. Zhu, X. Liu, and E. Zhu, "Oversampling with reliably expanding minority class regions for imbalanced data learning," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 6, pp. 6167–6181, Jun. 2023.

[20] Y. Xie, M. Qiu, H. Zhang, L. Peng, and Z. Chen, "Gaussian distribution based oversampling for imbalanced data classification," *IEEE Trans. Knowl. Data Eng.*, vol. 34, no. 2, pp. 667–679, Feb. 2022.

[21] W. Pei et al., "DG-SMOTE : A distance-angle-based genetic synthetic minority over-sampling technique for unbalanced data learning," *IEEE Trans. Evol. Comput.*, vol. 29, no. 6, pp. 2641–2655, Dec. 2025, doi: 10.1109/TEVC.2024.3515485.

[22] N. A. Azhar, M. S. M. Pozi, A. M. Din, and A. Jatowt, "An investigation of SMOTE based methods for imbalanced datasets with data complexity analysis," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 7, pp. 6651–6672, Jul. 2023.

[23] L. Li, H. He, and J. Li, "Entropy-based sampling approaches for multi-class imbalanced problems," *IEEE Trans. Knowl. Data Eng.*, vol. 32, no. 11, pp. 2159–2170, Nov. 2020.

[24] Z. Chen et al., "Multi-expert genetic programming-based ensemble for long-tailed image classification," *IEEE Trans. Evol. Comput.*, early access, doi: 10.1109/TEVC.2026.3669207.

[25] J. Ju, L. Zhai, W. Ren, L. Wang, and Y. Ren, "An effective imbalanced jpeg steganalysis scheme based on adaptive cost-sensitive feature learning," *IEEE Trans. Knowl. Data Eng.*, vol. 34, no. 3, pp. 1038–1052, Mar. 2022.

[26] E. R. Fernandes, A. C. De Carvalho, and X. Yao, "Ensemble of classifiers based on multiobjective genetic sampling for imbalanced data," *IEEE Trans. Knowl. Data Eng.*, vol. 32, no. 6, pp. 1104–1115, Jun. 2020.

[27] K. Yang, Z. Yu, C. P. Chen, W. Cao, J. You, and H.-S. Wong, "Incremental weighted ensemble broad learning system for imbalanced data," *IEEE Trans. Knowl. Data Eng.*, vol. 34, no. 12, pp. 5809–5824, Dec. 2022.

[28] H.-H. Lee, S. Park, and J. Im, "Resampling approach for one-class classification," *Pattern Recognit.*, vol. 143, Nov. 2023, Art. no. 109731.

[29] O. Miho, P. Wang, K. Matsuda, S. Katagiri, H. Watanabe, and A. Ralescu, "Confusion-matrix-based kernel logistic regression for imbalanced data classification," *IEEE Trans. Knowl. Data Eng.*, vol. 29, no. 9, pp. 1806–1819, Sep. 2017.

[30] M. Wasikowski and X.-W. Chen, "Combating the small sample class imbalance problem using feature selection," *IEEE Trans. Knowl. Data Eng.*, vol. 22, no. 10, pp. 1388–1400, Oct. 2010.

[31] R. Jiao, B. Xue, and M. Zhang, "Learning to preselection: A filter-based performance predictor for multiobjective feature selection in classification," *IEEE Trans. Evol. Comput.*, vol. 30, no. 1, pp. 31–45, Feb. 2026.

[32] X. Cai and Y. Xue, "A population initialization method based on similarity and mutual information in evolutionary algorithm for bi-objective feature selection," *ACM Trans. Evol. Learn.*, vol. 4, no. 3, pp. 1–21, 2024.

[33] E. Hancer, B. Xue, and M. Zhang, "A multimodal multi-objective evolutionary algorithm for filter feature selection in multi-label classification," *IEEE Trans. Artif. Intell.*, vol. 5, no. 9, pp. 4428–4442, Sep. 2024.

[34] F. Cheng, R. Zhang, Z. Huang, J. Qiu, M. Xia, and L. Zhang, "An objective space constraint-based evolutionary method for high-dimensional feature selection [Research Frontier]," *IEEE Comput. Intell. Mag.*, vol. 19, no. 2, pp. 113–128, May 2024.

[35] S. Han, K. Zhu, M. Zhou, H. Alhumade, and A. Abusorrah, "Locating multiple equivalent feature subsets in feature selection for imbalanced classification," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 9, pp. 9195–9209, Sep. 2023.

[36] H. Saadatmand and M. Akbarzadeh-T, "Many-objective Jaccard-based evolutionary feature selection for high-dimensional imbalanced data classification," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 12, pp. 8820–8835, Dec. 2024, doi: 10.1109/TPAMI.2024.3416196.

[37] Y. Zhang and Q. Yang, "A survey on multi-task learning," *IEEE Trans. Knowl. Data Eng.*, vol. 34, no. 12, pp. 5586–5609, Dec. 2021.

[38] K. C. Tan, L. Feng, and M. Jiang, "Evolutionary transfer optimization - A new frontier in evolutionary computation research," *IEEE Comput. Intell. Mag.*, vol. 16, no. 1, pp. 22–33, Feb. 2021.

[39] X. Song et al., "Evolutionary computation for feature selection in classification: A comprehensive survey of solutions, applications and challenges," *Swarm Evol. Comput.*, vol. 90, Oct. 2024, Art. no. 101661.

[40] K. Chen, B. Xue, M. Zhang, and F. Zhou, "An evolutionary multitasking-based feature selection method for high-dimensional classification," *IEEE Trans. Cybern.*, vol. 52, no. 7, pp. 7172–7186, Jul. 2022.

[41] N. Zhang, A. Gupta, Z. Chen, and Y.-S. Ong, "Evolutionary machine learning with minions: A case study in feature selection," *IEEE Trans. Evol. Comput.*, vol. 26, no. 1, pp. 130–144, Feb. 2022.

[42] X. Wang, H. Shangguan, F. Huang, S. Wu, and W. Jia, " MEL : Efficient multi-task evolutionary learning for high-dimensional feature selection," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 8, pp. 4020–4033, Aug. 2024.

[43] R. Jiao, B. Xue, and M. Zhang, "Benefiting from single-objective feature selection to multiobjective feature selection: A multiform approach," *IEEE Trans. Cybern.*, vol. 53, no. 12, pp. 7773–7786, Dec. 2023.

[44] J. Liang, Y. Zhang, B. Qu, K. Chen, K. Yu, and C. Yue, "A multiform optimization framework for multi-objective feature selection in classification," *IEEE Trans. Evol. Comput.*, vol. 28, no. 4, pp. 1024–1038, Aug. 2024.

[45] Y. Hu, Y. Zhang, D. Gong, and X. Sun, "Multiparticipant federated feature selection algorithm with particle swarm optimization for imbalanced data under privacy protection," *IEEE Trans. Artif. Intell.*, vol. 4, no. 5, pp. 1002–1016, Oct. 2023.

[46] H. Xu, B. Xue, and M. Zhang, "A duplication analysis-based evolutionary algorithm for biobjective feature selection," *IEEE Trans. Evol. Comput.*, vol. 25, no. 2, pp. 205–218, Apr. 2021.

[47] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan, "A fast and elitist multiobjective genetic algorithm: NSGA-II," *IEEE Trans. Evol. Comput.*, vol. 6, no. 2, pp. 182–197, Apr. 2002.

[48] R. Jiao, B. Xue, and M. Zhang, "A tri-objective method for bi-objective feature selection in classification," *Evol. Comput. (MIT Press)*, vol. 32, no. 3, pp. 217–248, 2024.

[49] R. Jiao, B. Xue, and M. Zhang, "Solving multiobjective feature selection problems in classification via problem reformulation and duplication handling," *IEEE Trans. Evol. Comput.*, vol. 28, no. 4, pp. 846–860, Aug. 2024.

[50] B. Xue, M. Zhang, and W. N. Browne, "Particle swarm optimization for feature selection in classification: A multi-objective approach," *IEEE Trans. Cybern.*, vol. 43, no. 6, pp. 1656–1671, Dec. 2013.

[51] Y. Lu, Y. Zhang, B. Han, Y.-M. Cheung, and H. Wang, "Label-noise learning with intrinsically long-tailed data," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.,* 2023, pp. 1369–1378.



**Ruwang Jiao** (Senior Member, IEEE) received the Ph.D. degree from the School of Mechanical Engineering and Electronic Information, China University of Geosciences, Wuhan, China, in 2020.

He is currently a Professor with Soochow University, Suzhou, China. His research mainly focuses on feature selection, dimensionality reduction, and evolutionary multiobjective learning.

Prof. Jiao serves as the Chair of the Task Force on Evolutionary Computation for Feature Selection and Construction within the IEEE Computational Intelligence Society (CIS).



**Naoki Masuyama** (Member, IEEE) received the B.Eng. degree in aerospace engineering from Nihon University, Funabashi, Japan, in 2010, the M.E. degree in human mechatronics systems from Tokyo Metropolitan University, Hino, Japan, in 2012, and the Ph.D. degree in computer science from the Faculty of Computer Science and Information Technology, University of Malaya, Kuala Lumpur, Malaysia, in 2016.

He is currently an Associate Professor with the Department of Core Informatics, Graduate School of Informatics, Osaka Metropolitan University, Osaka, Japan. His research interests include clustering, data mining, and continual learning.



**Yusuke Nojima** (Member, IEEE) received the B.S. and M.S. degrees in mechanical engineering from Osaka Institute of Technology, Osaka, Japan, in 1999 and 2001, respectively, and the Ph.D. degree in engineering from the Department of System Function Science, Kobe University, Hyogo, Japan, in 2004.

Since 2004, he has been with Osaka Prefecture University, Osaka, where he has been a Professor with the Department of Computer Science and Intelligent Systems, since 2020. Since 2022, he has been a Professor with the Department of Core Informatics, Graduate School of Informatics, Osaka Metropolitan University, Osaka. His research interests include evolutionary fuzzy systems, evolutionary multiobjective optimization, and multiobjective data mining.

Dr. Nojima was a Guest Editor for several special issues in international journals. He is a member of Fuzzy Systems Technical Committee of IEEE Computational Intelligence Society. He also serves as the President-Elect of International Fuzzy Systems Association (IFSA).